SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO3: Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 01 **Total Marks:100**

Annexure A

CASE STUDY

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Levels	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly		
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts		
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning		
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided		
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand		

Annexure A

Case Study 1: E-Commerce Customer Analysis
Scenario:
An e-commerce company has collected customer transaction data, including purchase amount, number of orders, browsing time, and customer ratings. The company wants to understand customer purchasing behavior and improve its marketing strategies.
Questions:
A. How can descriptive statistics such as mean, median, standard deviation, and quartiles be used to summarize customer transaction data?
B. Discuss how unusually high or low purchase amounts can affect the analysis and describe suitable methods for detecting outliers.
C. How can hypothesis testing be used to determine whether premium members spend significantly more than regular customers? Formulate the hypotheses and explain the analysis process.

Case Study 2: Renewable Energy Production Analysis
Scenario:
A solar power company has collected data from multiple solar plants, including daily energy generation, sunlight hours, panel temperature, and system efficiency. The company wants to evaluate plant performance and verify whether the average energy production meets the expected target.
Questions:
A. Explain how descriptive statistics and distribution analysis can be used to summarize and evaluate the energy production dataset.
B. Discuss the impact of abnormal energy generation values on the analysis and describe methods for detecting and handling outliers.
C. How can confidence intervals and hypothesis testing be used to determine whether the average daily energy generation differs significantly from the expected target value?

Case Study 3: Hospital Patient Recovery Analysis
Scenario:
A hospital has collected patient recovery data, including age, treatment cost, number of hospital days, recovery time, and patient satisfaction score. The hospital wants to analyze treatment effectiveness and improve patient care quality.
Questions:
A. How can descriptive statistics such as mean, median, standard deviation, and quartiles be used to summarize patient recovery time and treatment cost?
B. Discuss how unusually high hospital stay duration or treatment cost values may affect the analysis. What methods can be used to detect outliers?
C. How can hypothesis testing be used to determine whether patients receiving a new treatment recover faster than patients receiving the existing treatment? Formulate the hypotheses and explain the analysis process.

Case Study 4: Student Academic Performance Analysis
Scenario:
A university has collected student performance data, including attendance percentage,

assignment marks, internal test scores, study hours, and final exam marks. The university wants to identify factors affecting student success.

Questions:

A. Explain how descriptive statistics and distribution analysis can be used to summarize student marks and attendance data.

B. Discuss how extremely low or high marks can affect the interpretation of academic performance. Describe suitable methods for detecting and handling outliers.

C. How can hypothesis testing be used to check whether students with attendance above 75% score significantly higher than students with attendance below 75%? Formulate the hypotheses and explain the steps.

Case Study 5: Banking Loan Approval Analysis

Scenario:

A bank has collected customer loan application data, including income, credit score, loan amount, repayment history, employment status, and loan approval status. The bank wants to understand customer risk and improve loan approval decisions.

Questions:

A. How can descriptive statistics be used to summarize customer income, credit scores, and loan amounts?

B. Discuss how unusually high loan amounts or very low credit scores can influence the analysis. Explain methods for identifying such outliers.

C. How can hypothesis testing be used to determine whether customers with higher credit scores are more likely to receive loan approval than customers with lower credit scores? Formulate the hypotheses and describe the analysis process.

Case Study 6: Online Learning Platform Analysis

Scenario:

An online learning platform has collected learner activity data, including course completion percentage, quiz scores, video watch time, login frequency, and feedback ratings. The platform wants to improve course engagement and learner success.

Questions:

A. Explain how descriptive statistics can be used to summarize quiz scores, watch time, and course completion rates.

B. Discuss how unusually high or low learning activity values may affect the analysis. What techniques can be used to detect outliers?

C. How can hypothesis testing be used to determine whether learners who watch more than 70% of the course videos score significantly higher in quizzes than other learners? Formulate the hypotheses and explain the testing procedure.

Case Study 7: Retail Sales Performance Analysis

Scenario:

A retail chain has collected sales data from different branches, including daily sales amount, number of customers, discount percentage, product category, and customer feedback score. The company wants to evaluate branch performance and improve sales planning.

Questions:

A. How can descriptive statistics and distribution analysis be used to summarize daily sales and customer count?

B. Discuss how abnormal sales values, such as festival-season sales or very low sales days, can affect the analysis. Explain suitable outlier detection methods.

C. How can confidence intervals and hypothesis testing be used to determine whether the average sales of urban branches are significantly higher than rural branches?

Case Study 8: Food Delivery Service Analysis
Scenario:
A food delivery company has collected order data, including delivery time, order value, customer rating, distance travelled, and delivery partner experience. The company wants to reduce delivery delays and improve customer satisfaction.
Questions:
A. Explain how descriptive statistics such as mean, median, standard deviation, and quartiles can be used to analyze delivery time and customer ratings.
B. Discuss how unusually long delivery times can affect customer satisfaction analysis. Describe suitable methods for detecting outliers.
C. How can hypothesis testing be used to determine whether experienced delivery partners complete deliveries faster than new delivery partners? Formulate the hypotheses and explain the analysis process.

Case Study 9: Manufacturing Defect Analysis
Scenario:
A manufacturing company has collected production data, including number of units produced, defect count, machine temperature, production time, and operator shift. The company wants to reduce defects and improve product quality.
Questions:
A. How can descriptive statistics be used to summarize defect counts, production time, and machine temperature?
B. Discuss how abnormal defect values may affect the quality analysis. What methods can be used to detect outliers in production data?
C. How can hypothesis testing be used to determine whether the defect rate differs significantly between day shift and night shift production? Formulate the hypotheses and explain the analysis process.

Case Study 10: Mobile Network Performance Analysis
Scenario:
A telecom company has collected mobile network data, including signal strength, internet speed, call drop rate, number of users, and customer complaints from different locations. The company wants to improve network quality.
Questions:
A. Explain how descriptive statistics and distribution analysis can be used to summarize internet speed, signal strength, and call drop rate.
B. Discuss how extremely low internet speed or high call drop rate values can affect the analysis. Describe suitable outlier detection and handling methods.
C. How can confidence intervals and hypothesis testing be used to determine whether the average internet speed in urban areas is significantly different from rural areas?

Case Study 11: Public Transportation Usage Analysis
Scenario:
A city transport department has collected bus usage data, including passenger count, ticket revenue, waiting time, travel duration, and route distance. The department wants to improve transport scheduling and passenger service.

Questions:
A. How can descriptive statistics be used to summarize passenger count, waiting time, and travel duration?
B. Discuss how unusually high passenger counts during peak hours or unusually long waiting times can affect the analysis. Explain methods for detecting outliers.
C. How can hypothesis testing be used to determine whether average waiting time during peak hours is significantly higher than during non-peak hours? Formulate the hypotheses and explain the testing process.

Case Study 12: Agriculture Crop Yield Analysis
Scenario:
An agriculture department has collected crop production data from different farms, including rainfall, fertilizer usage, soil moisture, temperature, and crop yield. The department wants to analyze factors affecting crop productivity.
Questions:
A. Explain how descriptive statistics and distribution analysis can be used to summarize crop yield, rainfall, and fertilizer usage.
B. Discuss how abnormal crop yield values caused by drought, pests, or excessive rainfall can affect the analysis. Describe suitable outlier detection methods.
C. How can confidence intervals and hypothesis testing be used to determine whether farms using organic fertilizer have significantly different crop yield compared with farms using chemical fertilizer?

Case Study 13: Employee Productivity Analysis
Scenario:
A software company has collected employee productivity data, including working hours, number of completed tasks, project deadlines, experience level, and performance rating. The company wants to improve employee performance and project planning.
Questions:
A. How can descriptive statistics such as mean, median, standard deviation, and quartiles be used to summarize employee productivity data?
B. Discuss how unusually high or low task completion values can affect productivity analysis. Describe suitable methods for detecting outliers.
C. How can hypothesis testing be used to determine whether experienced employees complete significantly more tasks than newly joined employees? Formulate the hypotheses and explain the analysis process.

Case Study 14: Automobile Fuel Efficiency Analysis
Scenario:
An automobile company has collected vehicle performance data, including fuel consumption, engine capacity, vehicle weight, mileage, and maintenance cost. The company wants to analyze fuel efficiency and improve vehicle design.
Questions:
A. Explain how descriptive statistics and distribution analysis can be used to summarize mileage and fuel consumption data.
B. Discuss how abnormal mileage values can affect the analysis. What methods can be used to detect and handle outliers?
C. How can confidence intervals and hypothesis testing be used to determine whether hybrid cars have significantly better mileage than petrol cars?

Case Study 15: Social Media Engagement Analysis
Scenario:
A digital marketing agency has collected social media data, including number of likes, shares, comments, followers, posting time, and engagement rate. The agency wants to improve content strategy and audience reach.
Questions:
A. How can descriptive statistics be used to summarize likes, shares, comments, and engagement rate?
B. Discuss how viral posts with extremely high engagement values may affect the analysis. Describe suitable outlier detection methods.
C. How can hypothesis testing be used to determine whether video posts receive significantly higher engagement than image posts? Formulate the hypotheses and explain the testing procedure.

Case Study 16: Airline Passenger Satisfaction Analysis
Scenario:
An airline company has collected passenger feedback data, including ticket price, flight delay time, baggage waiting time, seat comfort rating, and overall satisfaction score. The airline wants to improve service quality.
Questions:
A. Explain how descriptive statistics can be used to summarize delay time, baggage waiting time, and passenger satisfaction scores.
B. Discuss how unusually long flight delays can affect the analysis of passenger satisfaction. What methods can be used to detect outliers?
C. How can hypothesis testing be used to determine whether passengers on on-time flights give significantly higher satisfaction ratings than passengers on delayed flights?

Case Study 17: Smart City Traffic Analysis
Scenario:
A smart city traffic department has collected traffic data, including vehicle count, average speed, signal waiting time, accident count, and pollution level from different roads. The department wants to reduce congestion and improve road safety.
Questions:
A. How can descriptive statistics and distribution analysis be used to summarize vehicle count, waiting time, and average speed?
B. Discuss how abnormal traffic values during accidents, festivals, or roadblocks can affect the analysis. Describe suitable methods for detecting outliers.
C. How can hypothesis testing be used to determine whether average waiting time is significantly higher during peak hours than during non-peak hours?

Case Study 18: Hotel Booking and Customer Review Analysis
Scenario:
A hotel chain has collected booking data, including room price, length of stay, customer rating, booking cancellation status, and service feedback score. The hotel wants to improve customer satisfaction and pricing strategy.

Questions:
A. How can descriptive statistics be used to summarize room price, length of stay, and customer ratings?
B. Discuss how unusually high room prices or extremely low customer ratings can affect the analysis. Explain suitable methods for detecting outliers.
C. How can hypothesis testing be used to determine whether customers who stay more than three days give significantly different ratings compared with customers who stay for one or two days?

Case Study 19: Fitness App User Health Analysis
Scenario:
A fitness app has collected user health data, including daily steps, calories burned, workout duration, heart rate, sleep hours, and fitness goal completion. The app wants to study user activity patterns and improve health recommendations.
Questions:
A. Explain how descriptive statistics can be used to summarize daily steps, calories burned, workout duration, and sleep hours.
B. Discuss how unusually high workout duration or abnormal heart rate values can affect the health analysis. What methods can be used to detect outliers?
C. How can hypothesis testing be used to determine whether users who sleep at least seven hours per day burn significantly more calories than users who sleep less than seven hours?

Case Study 20: Online Examination Performance Analysis
Scenario:
An educational institution has collected online exam data, including exam score, time taken to complete the test, number of attempts, internet interruption count, and student feedback. The institution wants to evaluate student performance and improve the online assessment system.
Questions:
A. How can descriptive statistics and distribution analysis be used to summarize exam scores, completion time, and number of attempts?
B. Discuss how unusually short completion time or extremely high number of attempts can affect exam performance analysis. Describe suitable outlier detection methods.
C. How can hypothesis testing be used to determine whether students with fewer internet interruptions score significantly higher than students with frequent interruptions?

ANSWERS:

Case Study 1: E-Commerce Customer Analysis

A. Descriptive Statistics

1. **Mean:** Calculates the average purchase amount of customers.
2. **Median:** Identifies the middle purchase value and helps understand typical customer spending.
3. **Standard Deviation:** Measures the variation in purchase amounts among customers.
4. **Quartiles:** Divide transaction data into four equal parts to understand different customer spending levels.

5. **Use:** These statistics help the company understand purchasing behavior and improve marketing strategies.

B. Outlier Detection

1. Unusually high or low purchase amounts can affect the accuracy of statistical summaries.
2. Very high purchase amounts may increase the average, while very low amounts may decrease it.
3. **Box Plots:** Help identify unusually high or low purchase values visually.
4. **IQR Method:** Identifies values that fall unusually far from the middle portion of the data.
5. **Z-Scores:** Help identify values that differ considerably from the average.
6. The company should investigate outliers and correct data-entry errors while retaining genuine transactions.

C. Hypothesis Testing

1. **Purpose:** Determines whether premium members spend more than regular customers.
2. **Null Hypothesis (H_0):** Premium and regular customers have equal average spending.
3. **Alternative Hypothesis (H_1):** Premium customers have higher average spending.
4. **Data Collection:** Collect purchase transaction data from both customer groups.
5. **Statistical Test:** Use an independent two-sample t-test to compare average spending.
6. **Decision:** Examine the p-value to determine whether the difference is statistically significant.
7. **Conclusion:** If the p-value is below the chosen significance level, reject the null hypothesis and conclude that premium members spend significantly more.

Case Study 3: Hospital Patient Recovery Analysis

A. Descriptive Statistics

1. **Mean:** Calculates the average patient recovery time and treatment cost.
2. **Median:** Identifies the middle recovery time and treatment cost.
3. **Standard Deviation:** Measures the variation in recovery time and treatment expenses among patients.
4. **Quartiles:** Divide patient data into four equal parts and help identify differences in recovery patterns.
5. **Use:** These statistics help hospitals evaluate treatment effectiveness and improve patient care.

B. Outlier Detection

1. Unusually long hospital stays or high treatment costs can affect statistical summaries.
2. These values may increase the average and misrepresent typical patient outcomes.
3. **Box Plots:** Visually identify unusually high treatment costs and long hospital stays.
4. **IQR Method:** Detects observations that are unusually far from the middle portion of the data.

5. **Z-Scores:** Identify values that differ considerably from the average.
6. Hospitals should investigate unusual observations and retain genuine patient records for accurate analysis.

C. Hypothesis Testing

1. **Purpose:** Determines whether patients receiving a new treatment recover faster than those receiving the existing treatment.
2. **Null Hypothesis (H_0):** The new treatment does not reduce average recovery time.
3. Alternative Hypothesis (H_1): The new treatment reduces average recovery time.
4. **Data Collection:** Collect recovery-time information from patients receiving both treatments.
5. **Statistical Test:** Use an appropriate independent two-sample t-test to compare recovery times.
6. **Decision:** Examine the p-value to determine whether the difference is statistically significant.
7. **Conclusion:** If the result is statistically significant and the new treatment group recovers faster, conclude that the new treatment is associated with reduced recovery time.

Case Study 5: Banking Loan Approval Analysis

A. Descriptive Statistics

1. **Mean:** Calculates the average customer income, credit score, and loan amount.
2. **Median:** Identifies the middle value and helps summarize financial data containing extreme values.
3. **Standard Deviation:** Measures variation in customer income, credit scores, and loan amounts.
4. **Quartiles:** Divide the financial data into four equal parts and help understand customer profiles.
5. **Use:** These statistics help banks assess customer financial characteristics and understand loan risk.

B. Outlier Detection

1. Unusually high loan amounts and very low credit scores can influence the analysis.
2. Extreme values may distort statistical summaries and affect risk assessment.
3. **Box Plots:** Help identify unusually high loan amounts and low credit scores.
4. **IQR Method:** Identifies values that are unusually distant from the middle portion of the data.
5. **Z-Scores:** Help identify values that differ considerably from the average.
6. Banks should verify unusual records before correcting errors or making loan decisions.

C. Hypothesis Testing

1. **Purpose:** Determines whether customers with higher credit scores are more likely to receive loan approval.
2. **Null Hypothesis (H_0):** Customers with higher credit scores are not more likely to receive loan approval than customers with lower credit scores.
3. **Alternative Hypothesis (H_1)**: Customers with higher credit scores are more likely to receive loan approval.
4. **Data Collection:** Collect credit scores and loan approval records from both customer groups.
5. **Statistical Test:** Compare approval rates using an appropriate two-proportion test or chi-square test.
6. **Decision:** Examine the p-value to determine whether the difference in approval rates is statistically significant.
7. **Conclusion:** A statistically significant result in the expected direction supports an association between higher credit scores and higher approval rates.

Case Study 7: Retail Sales Performance Analysis

A. Descriptive Statistics and Distribution Analysis

1. **Mean:** Calculates the average daily sales and customer count.
2. **Median:** Identifies typical daily sales and customer numbers.
3. **Standard Deviation:** Measures variation in sales and customer counts across different days.
4. **Quartiles:** Divide sales data into four equal parts and help understand sales distribution.
5. **Distribution Analysis:** Histograms and box plots help identify sales patterns, seasonal variations, and unusual values.
6. **Use:** These statistics help the company evaluate branch performance and improve sales planning.

B. Outlier Detection

1. Unusually high or low sales values can affect the accuracy of sales analysis.
2. Festival-season sales may increase the average, while very low sales days may decrease it.
3. **Box Plots:** Help identify unusually high or low sales values.
4. **IQR Method:** Detects sales observations that are unusually far from the middle portion of the data.
5. **Z-Scores:** Help identify sales values that differ considerably from the average.
6. Genuine seasonal sales should be retained and analyzed appropriately, while data-entry errors should be corrected.

C. Confidence Intervals and Hypothesis Testing

1. **Purpose:** Determines whether urban branches have significantly higher average sales than rural branches.
2. **Null Hypothesis (H_0):** Average sales of urban branches are less than or equal to those of rural branches.

3. **Alternative Hypothesis (H_1):** Average sales of urban branches are higher than those of rural branches.
4. **Data Collection:** Collect sales data from urban and rural branches.
5. **Confidence Interval:** Estimates the difference in average sales and indicates the uncertainty of the estimate.
6. **Statistical Test:** Use an independent two-sample t-test to compare average sales.
7. **Conclusion:** If the results are statistically significant and urban branches have higher average sales, conclude that urban branches perform better in terms of average sales.

Case Study 9: Manufacturing Defect Analysis

A. Descriptive Statistics

1. **Mean:** Calculates the average defect count, production time, and machine temperature.
2. **Median:** Identifies the middle value for each production measurement.
3. **Standard Deviation:** Measures variation in defect counts, production time, and machine temperature.
4. **Quartiles:** Divide the observations into four equal parts and help understand production data.
5. **Use:** These statistics help the company monitor production efficiency, machine performance, and product quality.

B. Outlier Detection

1. Abnormally high defect counts can affect the accuracy of quality analysis.
2. High defect values may indicate machine faults, poor raw materials, or production problems.
3. **Box Plots:** Help identify unusually high or low defect counts.
4. **IQR Method:** Identifies defect observations that are unusually far from the middle portion of the data.
5. **Z-Scores:** Help detect values that differ considerably from the average.
6. The company should investigate unusual observations, correct data errors, and address genuine production problems.

C. Hypothesis Testing

1. **Purpose:** Determines whether defect rates differ significantly between day-shift and night-shift production.
2. **Null Hypothesis (H_0):** The defect rates are equal for day-shift and night-shift production.
3. **Alternative Hypothesis (H_1):** The defect rates differ between the two shifts.
4. **Data Collection:** Collect the number of defective units and total units produced during both shifts.
5. **Statistical Test:** Use an appropriate two-proportion test or chi-square test to compare defect rates.
6. **Decision:** Examine the p-value to determine whether the difference is statistically significant.

7. **Conclusion:** If the result is statistically significant, conclude that the defect rates differ between the two shifts and investigate possible causes.

Case Study 11: Public Transportation Usage Analysis

A. Descriptive Statistics

1. **Mean:** Calculates the average passenger count, waiting time, and travel duration.
2. **Median:** Identifies typical waiting times and journey durations.
3. **Standard Deviation:** Measures variation in passenger demand and travel times.
4. **Quartiles:** Divide the data into four equal parts and help identify different passenger and waiting-time patterns.
5. **Use:** These statistics help transport departments understand passenger demand and improve transportation scheduling.

B. Outlier Detection

1. Unusually high passenger counts and long waiting times can affect statistical summaries.
2. High passenger counts may indicate overcrowding, while long waiting times may indicate insufficient bus frequency or traffic delays.
3. **Box Plots:** Help identify unusually high passenger counts and long waiting times.
4. **IQR Method:** Detects observations that are unusually far from the middle portion of the data.
5. **Z-Scores:** Help identify values that differ considerably from the average.
6. Genuine peak-hour observations should be retained and analyzed appropriately to improve transport planning.

C. Hypothesis Testing

1. **Purpose:** Determines whether average waiting time is significantly higher during peak hours than during non-peak hours.
2. **Null Hypothesis (H_0):** Average peak-hour waiting time is less than or equal to average non-peak-hour waiting time.
3. **Alternative Hypothesis (H_1):** Average peak-hour waiting time is higher than average non-peak-hour waiting time.
4. **Data Collection:** Collect waiting-time information from peak and non-peak periods.
5. **Statistical Test:** Use an appropriate independent two-sample t-test to compare average waiting times.
6. **Decision:** Examine the p-value to determine whether the difference is statistically significant.
7. **Conclusion:** If the result is statistically significant, conclude that peak-hour waiting times are higher and consider improving bus frequency and scheduling.

Case Study 13: Employee Productivity Analysis

A. Descriptive Statistics

1. **Mean:** Calculates the average working hours and number of completed tasks.
2. **Median:** Identifies the typical employee's task completion level.
3. **Standard Deviation:** Measures variation in productivity among employees.
4. **Quartiles:** Divide productivity data into four equal parts and help identify different employee performance levels.
5. **Use:** These statistics help organizations evaluate employee productivity and improve project planning.

B. Outlier Detection

1. Unusually high or low task completion values can distort average productivity.
2. High values may indicate exceptional performance, while low values may result from difficult assignments or incomplete records.
3. **Box Plots:** Help identify unusually high or low task completion values.
4. **IQR Method:** Detects observations that are unusually far from the middle portion of the data.
5. **Z-Scores:** Identify values that differ considerably from the average.
6. Organizations should verify unusual records and consider workload and task complexity before interpreting productivity results.

C. Hypothesis Testing

1. **Purpose:** Determines whether experienced employees complete significantly more tasks than newly joined employees.
2. Null Hypothesis (H_0): Experienced employees do not complete more tasks on average than newly joined employees.
3. Alternative Hypothesis (H_1): Experienced employees complete more tasks on average than newly joined employees.
4. Data Collection: Collect task completion information from both employee groups.
5. Statistical Test: Use an appropriate independent two-sample t-test to compare average task completion.
6. Decision: Examine the p-value to determine whether the difference is statistically significant.
7. Conclusion: If the result is statistically significant and experienced employees have higher average task completion, conclude that they complete more tasks on average.

Case Study 15: Social Media Engagement Analysis

A. Descriptive Statistics

1. Mean: Calculates the average number of likes, shares, comments, and engagement rate.
2. Median: Identifies typical engagement levels and reduces the influence of extremely popular posts.
3. Standard Deviation: Measures variation in engagement across social media posts.

4. Quartiles: Divide engagement data into four equal parts and help identify different engagement levels.
5. Use: These statistics help digital marketing agencies understand audience engagement and improve content strategies.

B. Outlier Detection

1. Viral posts with extremely high engagement can increase the average and distort the analysis.
2. These posts may make ordinary posts appear less engaging than they actually are.
3. Box Plots: Help identify posts with unusually high or low engagement.
4. IQR Method: Detects engagement values that are unusually far from the middle portion of the data.
5. Z-Scores: Help identify values that differ considerably from the average.
6. Genuine viral posts should be retained and analyzed separately when appropriate to understand their performance.

C. Hypothesis Testing

1. Purpose: Determines whether video posts receive significantly higher engagement than image posts.
2. Null Hypothesis (H_0): Video posts and image posts have equal average engagement.
3. Alternative Hypothesis (H_1): Video posts have higher average engagement than image posts.
4. Data Collection: Collect engagement data for video and image posts.
5. Statistical Test: Use an appropriate independent two-sample t-test to compare average engagement.
6. Decision: Examine the p-value to determine whether the difference is statistically significant.
7. Conclusion: If the result is statistically significant and video posts have higher average engagement, conclude that video posts receive higher engagement.

Case Study 17: Smart City Traffic Analysis

A. Descriptive Statistics and Distribution Analysis

1. Mean: Calculates the average vehicle count, signal waiting time, and vehicle speed.
2. Median: Identifies typical traffic conditions and reduces the influence of extreme observations.
3. Standard Deviation: Measures variation in traffic volume, speed, and waiting time.
4. Quartiles: Divide traffic data into four equal parts and help identify different traffic patterns.
5. Distribution Analysis: Histograms and box plots help identify congestion patterns and unusual traffic conditions.
6. Use: These statistics help traffic departments understand road conditions and improve traffic management.

B. Outlier Detection

1. Accidents, festivals, and roadblocks can produce abnormal traffic values.
2. These events may cause unusually high congestion, long waiting times, or very low vehicle speeds.
3. Box Plots: Help identify unusually high or low traffic observations.
4. IQR Method: Detects traffic values that are unusually far from the middle portion of the data.
5. Z-Scores: Help identify values that differ considerably from the average.
6. Genuine incident-related observations should be retained and analyzed appropriately to understand their impact on traffic flow.

C. Hypothesis Testing

1. Purpose: Determines whether average waiting time is significantly higher during peak hours than during non-peak hours.
2. Null Hypothesis (H_0): Average peak-hour waiting time is less than or equal to average non-peak-hour waiting time.
3. Alternative Hypothesis (H_1): Average peak-hour waiting time is higher than average non-peak-hour waiting time.
4. Data Collection: Collect traffic waiting-time data during peak and non-peak hours.
5. Statistical Test: Use an appropriate independent two-sample t-test to compare average waiting times.
6. Decision: Examine the p-value to determine whether the difference is statistically significant.
7. Conclusion: If the result is statistically significant, conclude that waiting time is higher during peak hours and recommend suitable traffic management improvements.

Case Study 19: Fitness App User Health Analysis

A. Descriptive Statistics

1. Mean: Calculates the average daily steps, calories burned, workout duration, and sleep hours.
2. Median: Identifies typical daily activity and sleep patterns.
3. Standard Deviation: Measures variation in physical activity and sleep duration among users.
4. Quartiles: Divide the observations into four equal parts and help identify different activity levels.
5. Use: These statistics help fitness applications understand user behavior and provide personalized fitness recommendations.

B. Outlier Detection

1. Unusually high workout durations or abnormal heart-rate values can affect health data analysis.
2. These values may result from sensor errors, unusual activities, or genuine individual differences.
3. Box Plots: Help identify unusually high or low workout durations and heart-rate readings.

4. IQR Method: Detects observations that are unusually far from the middle portion of the data.
5. Z-Scores: Help identify values that differ considerably from the average.
6. The application should verify sensor accuracy and investigate unusual readings before excluding them. Potentially concerning heart-rate values should be handled carefully.

C. Hypothesis Testing

1. Purpose: Determines whether users who sleep at least seven hours burn significantly more calories than users who sleep less than seven hours.
2. Null Hypothesis (H_0): Users who sleep at least seven hours do not burn more calories on average than users who sleep less than seven hours.
3. Alternative Hypothesis (H_1): Users who sleep at least seven hours burn more calories on average than users who sleep less than seven hours.
4. Data Collection: Collect sleep-duration and calorie-expenditure data from both groups.
5. Statistical Test: Use an appropriate independent two-sample t-test to compare average calories burned.
6. Decision: Examine the p-value to determine whether the difference is statistically significant.
7. Conclusion: If the result is statistically significant and the longer-sleep group burns more calories on average, conclude that the groups differ in calorie expenditure. This association does not necessarily mean that longer sleep directly causes higher calorie expenditure.